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LIBRARYSYS PROJECT MANUAL

SERVER-SIDE DEVELOPMENT

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LIBRARY MANAGEMENT SYSTEM **21**

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AUTHOR *21*
STUDENT ID *21*
INSTITUTION *21*
YEAR *21*

1. Project Overview

LibrarySYS is an intuitive library management system designed to support the daily operation of a library. The system is designed to streamline book management, member management, loan transaction and return management, and performing administration, while ensuring data integrity and strict adherence to the rules of the business.

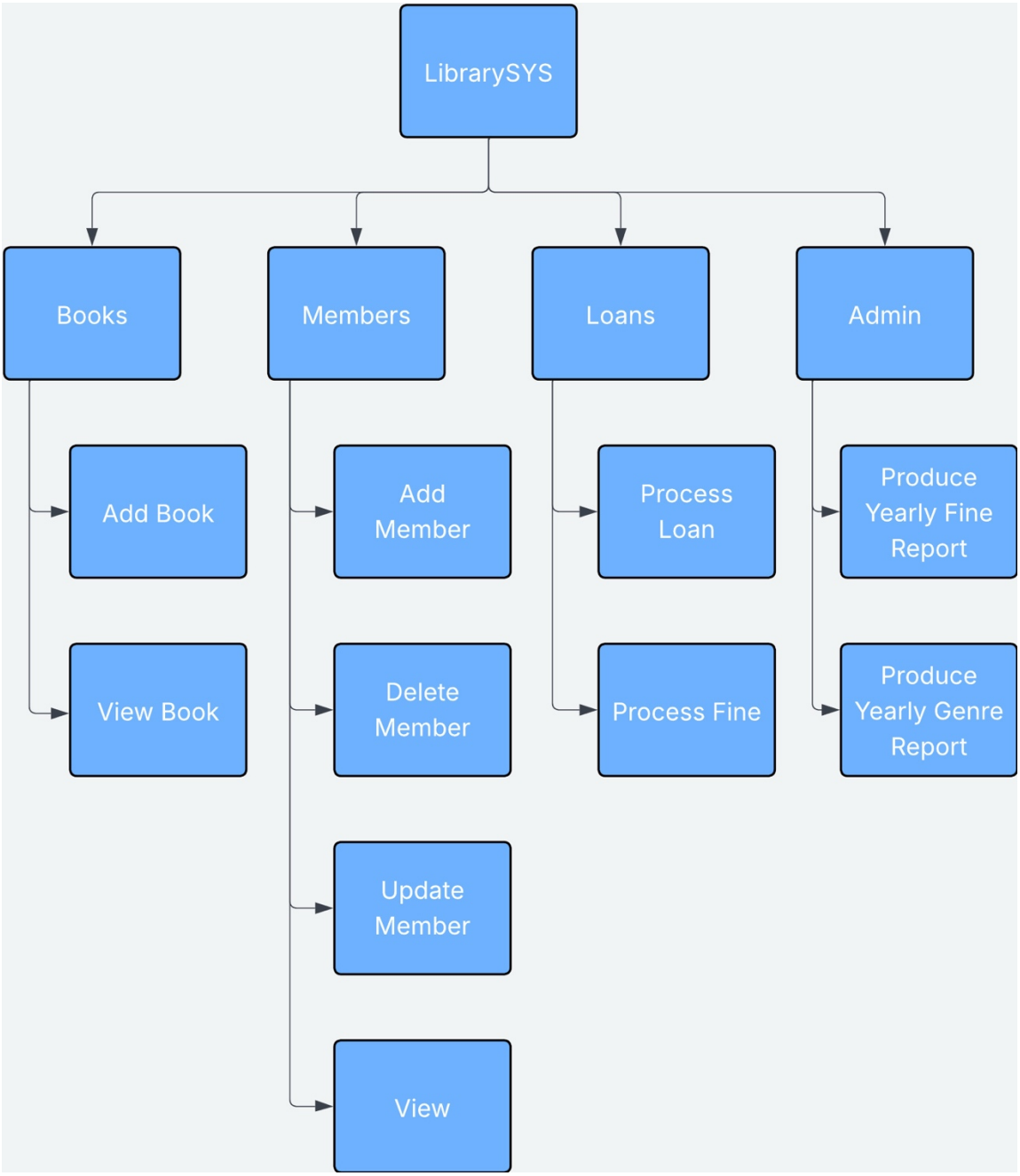
The system supports three levels of user access: librarian, reception, and manager, with each of these users having varying levels of permissions. A librarian can manage books, reception can manage books, members and process loans, and the manager is granted full access to all areas of the system including administrative reporting.

The core functionality of the system includes:

- **Book Management** – grants staff the ability to add and view books in the catalogue. All books in the library system are identified by ISBN and are tracked by the availability status.
- **Member Management** – grants staff the ability to register new members, update their details, view all member records, and change the status of a member.
- **Loan Processing** – grants staff the ability to process a member loan in a structured workflow, including validation checking and receipt generation.
- **Receipt Generation** – upon successful processing of a loan, the system automatically generates a PDF using the FPDF library. The receipt includes the members details, the books loaned, the loan and due date.

The system follows object-oriented architecture using a service and repository style. The business logic is handled by the library service, which acts as an intermediary between the user interface and the repository classes responsible for all database operations. This allows for effective separation of concerns and ensures a maintainable and scalable codebase.

2. Functional Components



3. Database Overview

Database Name: LibrarySYS

SQL File Name: LibrarySYS.sql

Database Schema: LibrarySYS

Relation Books

BookID TINYINT(3) AUTO_INCREMENT

Title VARCHAR(30) NOT NULL

Author VARCHAR(25) NOT NULL

Description VARCHAR(30) NOT NULL

ISBN CHAR(17) NOT NULL UNIQUE

Genre VARCHAR(20) NOT NULL

Publisher VARCHAR(25) NOT NULL

PublicationDate DATE

Status ENUM ('A', 'U') NOT NULL DEFAULT 'A'

Primary Key: BookID

Relation Members

MemberID TINYINT (3) AUTO_INCREMENT

FirstName VARCHAR (30) NOT NULL

LastName VARCHAR (30) NOT NULL

DOB Date NOT NULL

Phone VARCHAR (12) NOT NULL

Email VARCHAR (40) NOT NULL

AddressLine1 VARCHAR (30) NOT NULL

AddressLine2 VARCHAR (30)

City VARCHAR (30) NOT NULL

County VARCHAR (10) NOT NULL

Eircode CHAR (7) NOT NULL

RegistrationDate Date NOT NULL

Status ENUM ('A', 'I') DEFAULT 'A'

Primary Key: MemberID

Relation Loans

LoanID TINYINT (4) AUTO_INCREMENT

LoanDate Date NOT NULL

DueDate Date NOT NULL

MemberID TINYINT (3) NOT NULL

Primary Key: LoanID

Foreign Key: MemberID References Members (MemberID)

Relation LoanItems

LoanID TINYINT (4) NOT NULL

BookID TINYINT (3) NOT NULL

ReturnDate DATE

Primary Key: LoanID, BookID

Foreign Key: LoanID References Loans (LoanID)

BookID References Books (BookID)

Relation Fines

FineID TINYINT(4) AUTO_INCREMENT

FineAmount DECIMAL(5, 2) NOT NULL

Status ENUM ('U', 'P') NOT NULL DEFAULT 'U'

LoanID TINYINT(4) NOT NULL

BookID TINYINT(3) NOT NULL

Primary Key: FineID

Foreign Key: LoanID, BookID References LoanItems (LoanID, BookID)

Relation Users

UserID TINYINT(2) AUTO_INCREMENT

Username VARCHAR(30) NOT NULL UNIQUE

Password VARCHAR(255) NOT NULL

Primary Key: **UserID**

Database View

For the purposes of displaying the information contained in the tables in the library system, certain attributes were abbreviated. For correct naming, please refer to the above database schema.

Relation Books

ID	Title	Author	Description	ISBN	Genre	Publisher	Publication	Status
1	The Maze Runner 1	James Dashner	Dystopian Novel	978-0-385-73794-4	Science Fiction	Delacorte Press	2009-10-06	A
2	The Hunger Games	Suzanne Collins	Dystopian Novel	978-0-439-02352-8	Science Fiction	Scholastic Press	2008-09-14	A
3	Divergent	Veronica Roth	Science Fiction Novel	978-0-06-202403-9	Science Fiction	Katherine Tegen Books	2011-04-25	A
4	Harry Potter	J.K. Rowling	Fantasy Novel	978-0-7475-3269-9	Fantasy	Bloomsbury	1997-06-26	A
5	Percy Jackson	Rick Riordan	Fantasy Adventure	978-0-7868-5629-9	Fantasy	Disney Hyperion	2005-06-28	A
6	Dune	Frank Herbert	A desert planet	978-0-4410-1359-3	Science Fiction	Chilton Books	1965-08-01	A
7	The Hobbit	J.R.R. Tolkien	A hobbit adventure	978-0-6180-0221-4	Fantasy	Allen and Unwin	1937-09-21	A
8	The Shining	Stephen King	Haunted hotel	978-0-3850-0751-1	Horror	Doubleday	1977-01-28	A
9	Dracula	Bram Stoker	Classic vampire	978-0-1430-5459-7	Horror	Archibald Constable	1897-05-26	A
10	Gone Girl	Gillian Flynn	A vanishing wife	978-0-3075-8836-4	Thriller	Crown Publishing	2012-06-05	A

Relation Members

ID	FName	LName	DOB	Phone	Email	Add1	Add2	City	County	Eircode	Reg	Status
1	Dan	Courtney	2004-02-12	0874032491	dancourtney@gmail.com	14 Rockfeller Road	Rockfeller Lane	Killorglin	Kerry	V93V2F8	2025-05-01	A
2	Sarah	Obrien	1999-05-23	0865127784	sarahobrien@gmail.com	7 Oakwood Drive	Oakwood Estate	Tralee	Kerry	V93A3F2	2025-06-01	A
3	Michael	Horgan	1985-11-11	0853349012	michael.horgan@yahoo.ie	22 Riverside View	Riverside	Listowel	Galway	V31K2X9	2025-09-01	A
4	Eoin	Mannix	2005-05-02	0872345678	eoin.mannix22@gmail.com	1 Orchard Way	Ballydribeen	Killarney	Kerry	V93H5X0	2026-01-01	A
5	Andrew	Johnson	2005-03-01	0873241529	drewJones13@gmail.com	1 Hemmington Way	Varlington Road	Drumcondra	Dublin	V24F568	2026-03-20	A
6	James	Murphy	1990-05-14	0871452684	James.murphy@gmail.com	12 Oak Street	Apt 3	Dublin	Dublin	D01AB23	2022-03-10	A
7	Emma	Walsh	1978-08-03	0874567890	emma.walsh@outlook.com	23 Birch Lane	Block B	Limerick	Limerick	V94GH89	2020-05-05	A
8	Patrick	Fitzgerald	1972-12-09	0878903286	pat.fitz@gmail.com	34 Spruce Walk	Ground Floor	Wexford	Wexford	Y35MN45	2021-02-28	A
9	Dua	Lipa	1982-01-01	0871273246	dula.peep@gmail.com	1 Angel Way	PatricksTown	Dundalk	Limerick	V93H792	2026-01-02	A
10	Jamie	Courtney	2009-04-24	0874567143	Jamiecourtney8@gmail.com	Ownagarry	-	Killorglin	Kerry	V93H756	2025-06-12	A

Relation Loans

LoanID	LoanDate	DueDate	MemberID
1	2025-06-01	2025-06-06	1
2	2025-07-10	2025-07-15	2
3	2025-10-20	2025-10-25	3
4	2026-01-10	2026-01-15	1
5	2026-02-20	2026-02-25	4
6	2026-03-20	2026-03-25	1
7	2026-03-29	2026-04-03	2
8	2026-03-09	2026-04-03	2

Relation LoanItems

LoanID	BookID	ReturnDate
1	1	2025-06-05
1	2	2025-06-05
2	4	2025-07-15
3	7	2025-10-21
4	5	2026-01-14
4	6	2026-01-22
4	3	2026-01-22
5	6	2026-03-01
6	1	2026-03-29
6	2	2026-03-29
6	3	2026-03-21
6	4	2026-03-26
6	5	2026-03-20
7	1	2026-03-29
8	2	2026-03-29

Relation Fines

FineID	FineAmount	Status	LoanID	BookID
1	1.40	U	4	6
2	1.40	U	4	3
3	0.80	P	5	6
4	0.20	U	6	4

Relation Users

- Please note the username and password for each user is the same.

UserID	Username	Password
1	manager	$2y$10$CGzrGchuHnB6t4ebQ936lOvT7WL9YW6YPnzGl6hSv9aVCJx3UCmji
2	librarian	$2y$10$6TUl9q1r8AMrTENPcT1cxeErhOOX3QzquX5oyZCiKsJ0COlPbqg/.
3	reception	$2y$10$T5gRir3zI.7NRHzwMCiSaewrkHdJcHuXsEfbdTikrRLzXG0JIq5ny

4. System Functionality

The **config.php** file is the central configuration file for the application and is included at the top of every page. This file is responsible for setting up the application environment prior to any page logic execution.

It performs the following tasks:

- **Database connection** – db.php establishes the connection to the MariaDB database. This singular connection is injected into all repository files for successful database connectivity.
- **Class files** – all model classes are included for availability throughout the application.
- **Session** – the session is called here so that session data such as the logged in user, the member being processed and the books in a loan cart persist across pages.
- **System constants** – constants are defined, which control key business rules of the system:
    - LOAN_LENDING_PERIOD – the number of consecutive days a member can borrow a book. The current period is 5 days.
    - MAX_BOOKS_PER_LOAN – the maximum number of books a member can borrow in a single loan. The maximum is currently set to 5.
    - FINE_RATE_PER_DAY – the daily fine rate charged for overdue books. The fine is currently set to €0.20 per day per item.
- **Repository and Service instantiation** – the four repository objects responsible for communicating with the database tables are created and injected into the service constructor. This means every page in the application utilizes a shared service instance, which handles all business logic and executes repository methods.

When navigating the library system, please note the menu bar to the left of the screen. This bar is to be used for all navigation between pages and provides the following functionality:

- **Menu toggle** – open and close the navigation sidebar, collapsing it to icons only to maximize screen space.
- **Logged in user** – view of currently logged in staff member.
- **Dashboard** – provides a link to the dashboard, displaying key system statistics and serving as the home page of the application.
- **Books** – opens a sub-menu when clicked, displaying links to the Add Book and View Book pages.
- **Members** – opens a sub-menu when clicked, displaying a link to the member management page where staff can add, update, view, and alter member status.
- **Loans** – opens a sub-menu when clicked, displaying links to the Process Loan and Process Fine pages.
- **Admin** – opens a sub-menu, when clicked, displaying links to Produce a Fine Report and Produce a Genre Report.

- **Logout** – logs the current user out of the system and redirects them to the login screen.
- **Light mode** – toggles the colour scheme of the application between light and dark mode, allowing staff to select their preferred display setting.

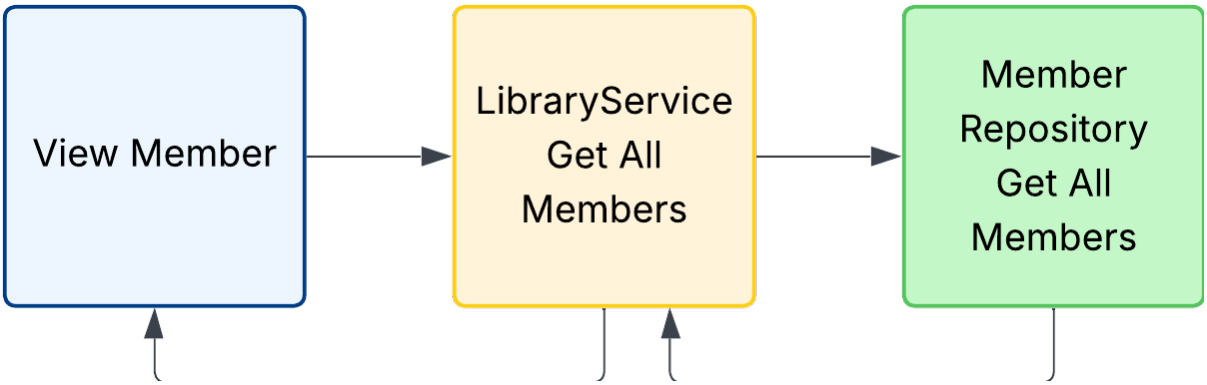
4.1. View

4.1.1. View All Books

- Navigate to the **Books** menu item in the left navigation panel.
- Click **View Books**.
- This brings you to **viewbook.php**.
- All books are retrieved from the books table and displayed.

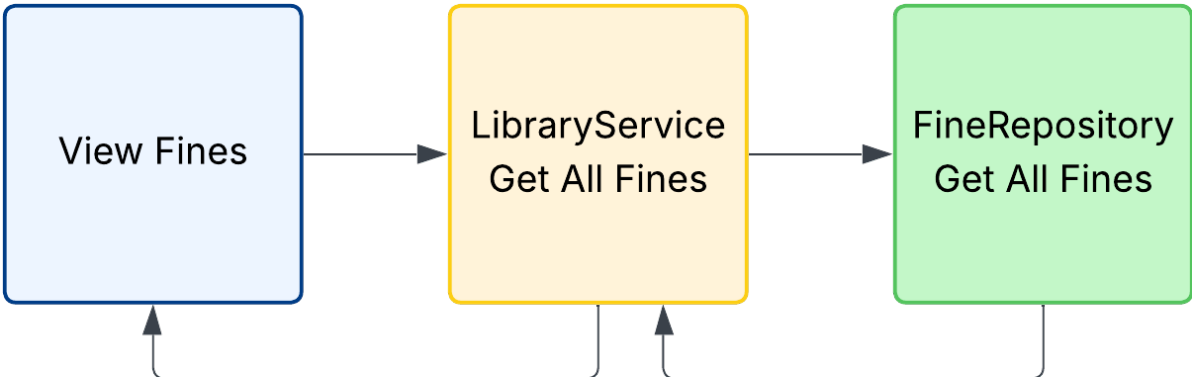
4.1.2 View Member

- Navigate to the **Members** menu item in the left navigation panel.
- Click **Update Members**.
- This brings you to **memberCRUD.php**.
- All members are retrieved from the members table and displayed.



4.1.2 View Fines

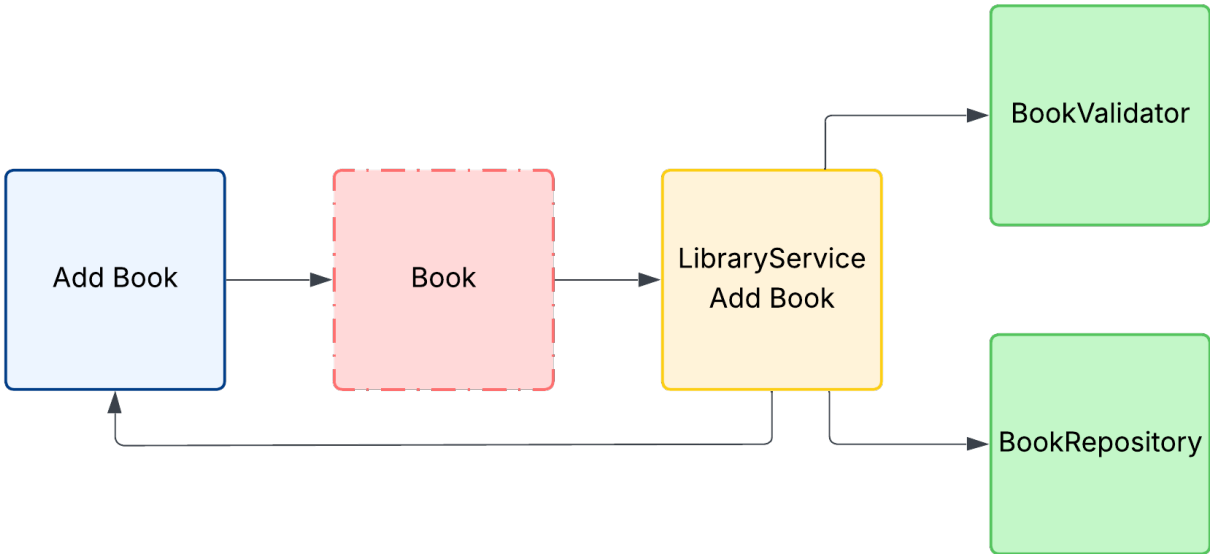
- Navigate to the **Loans** menu item in the left navigation panel.
- Click **Process Fine**.
- This brings you to **processFine.php**.
- All fines are retrieved from the fines table and displayed.



4.2. Insert

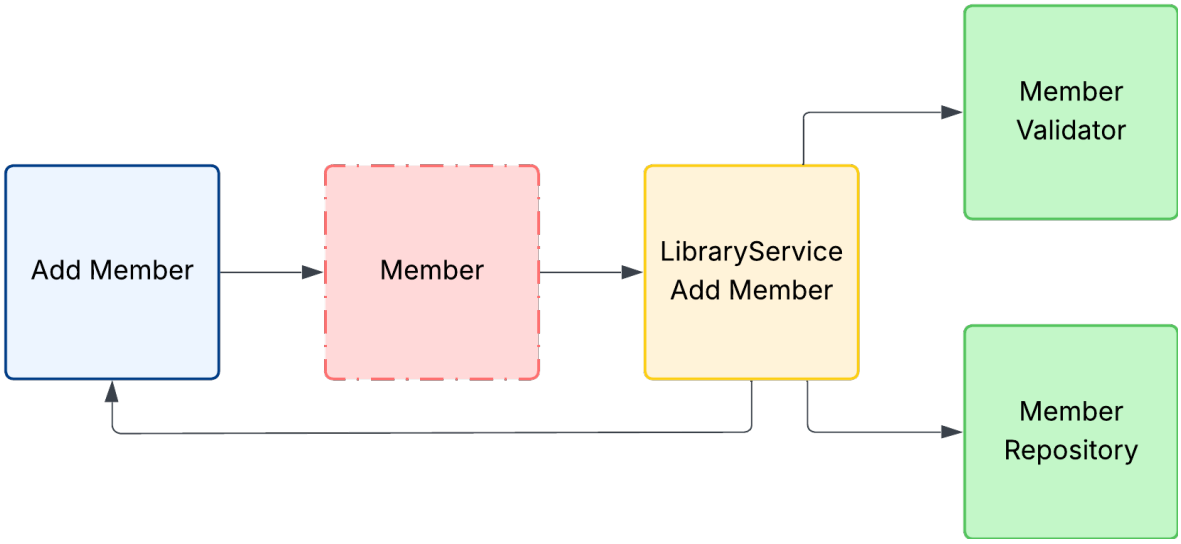
4.2.1. Add Book

- Navigate to the **Books** menu item in the left navigation panel.
- Click **Add Book**.
- This brings you to **addBook.php.**
- Enter all required fields for the book.
- Click **Add Book** to submit the details.
- The book is validated by the **BookValidator** and inserted into the Books table using the **BookRepository**.



4.2.1. Add Member

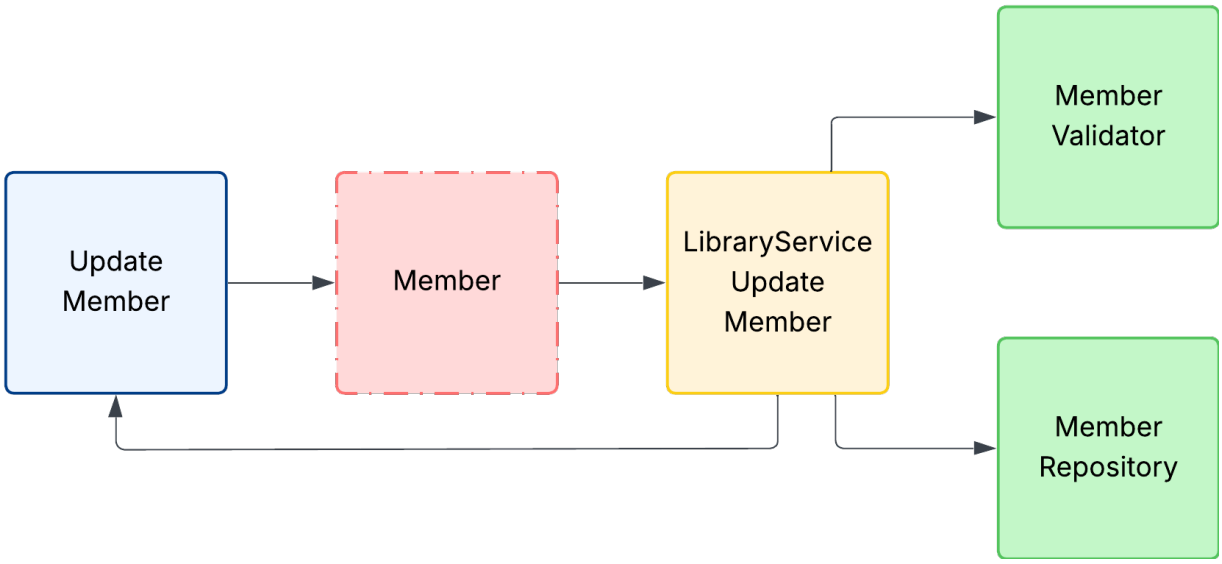
- Navigate to the **Member** menu item in the left navigation panel.
- Click **Update Members**.
- This brings you to **memberCRUD.php.**
- Click the **Add Member** button.
- Enter all required fields for the member.
- Click **Add Member** to submit the details.
- The member is validated by the **MemberValidator** and inserted into the Members table using the **MemberRepository**.



4.3. Update

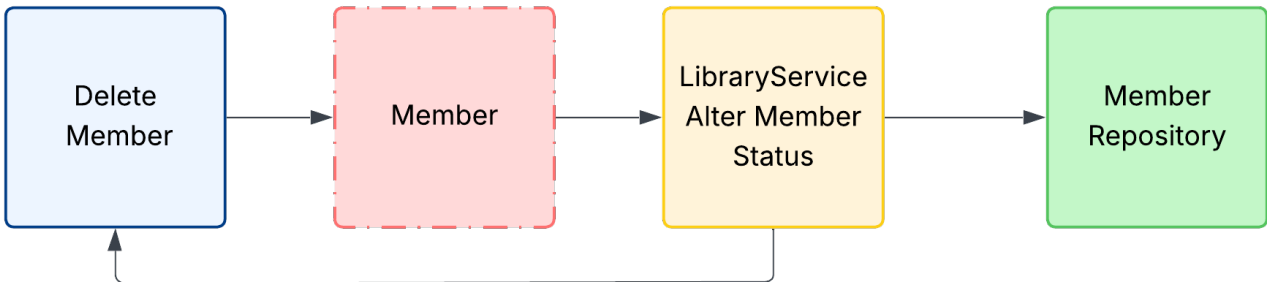
4.3.1. Update Member

- Navigate to the **Member** menu item in the left navigation panel.
- Click **Update Members**.
- Click the **Edit** button on any member's row.
- An edit form appears with the members details populated.
- Update the required fields and click **Update Member**.
- The updated details are validated by the **MemberValidator** and updated using the **MemberRepository**.



4.3.2. Update Member Status

- Navigate to the **Member** menu item in the left navigation panel.
- Click **Update Members**.
- Click the **Delete** button on any member's row.
- This toggles the status of the member from active to inactive and prevents loans being processed for inactive members.



4.3.2. Update Book Status

- Occurs during loan processing.
- When a loan is confirmed in **processLoan.php**, each book in the cart has its status changed from available to unavailable using the alter book status method in **BookRepository.**

4.4. Delete

4.4.1. Forgive Fine

- Navigate to the **Loans** menu item in the left navigation panel.
- Click **Process Fine**.
- This brings you to **processFine.php**.
- Click **Forgive Fine** button on any fine record.
- The fine record is hard deleted from the Fines table via the **Delete Fine** method in **FineRepository**.

4.5. Select

4.5.1. Process Loan - Get Unpaid Member Fines

- Occurs during loan processing.
- When a loan is being processed in **processLoan.php**, the member fines are checked to ensure no unpaid fines occur. In the event a fine is associated with a member, the member will be prevented from transacting. **(FineRepository - getUnpaidMemberFine).**

4.5.2. Process Loan – Has Overdue Books

- Occurs during loan processing.
- When a loan is being processed in **processLoan.php**, the members transaction history is reviewed to validate that no overdue books are associated with the member. If an overdue book is associated with a member, the member will be prevented from transacting. **(LoanRepository - hasOverdueBooks).**

4.5.3. Dashboard – Get Recent Loans

- Displayed on the dashboard page.
- When the page **index.php** is loaded, a method is called which retrieves the five most recent loans .
- This displays the book title, member name, loan date, due date, and status. (**LoanRepository - getRecentLoans).**

4.5.3. Dashboard – Get Top Borrowers

- Displayed on the dashboard page.
- When the page **index.php** is loaded, a method is called which retrieves the three top borrowers.
- This displays the member details as well as the count of loans. **(LoanRepository - getTopBorrowers).**

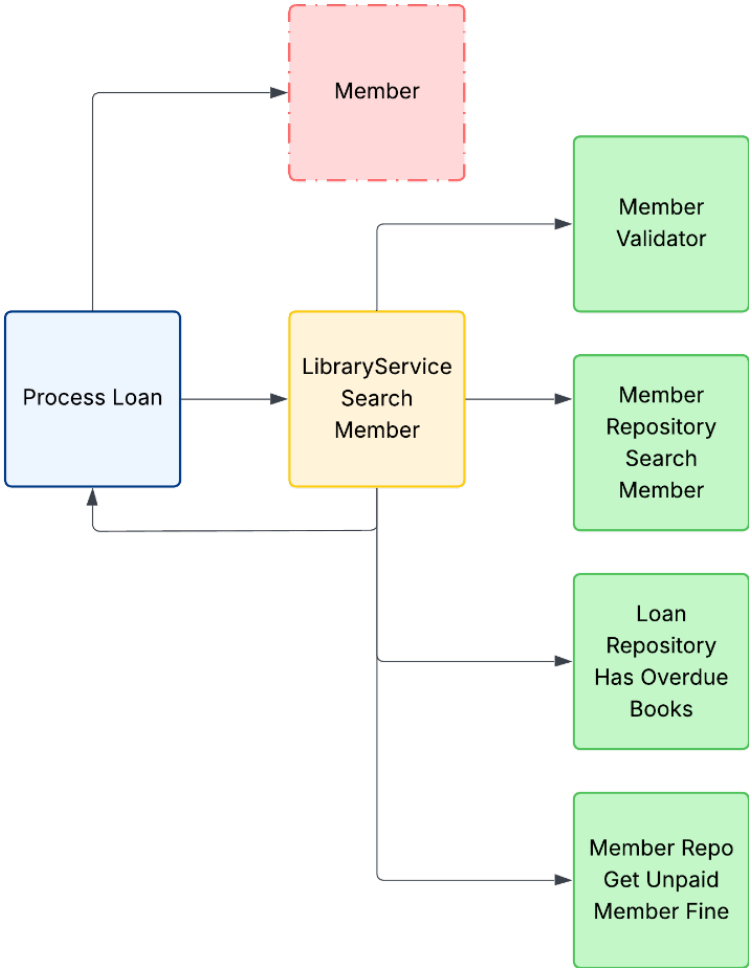
4.6. Process

4.6.1. Process Loan

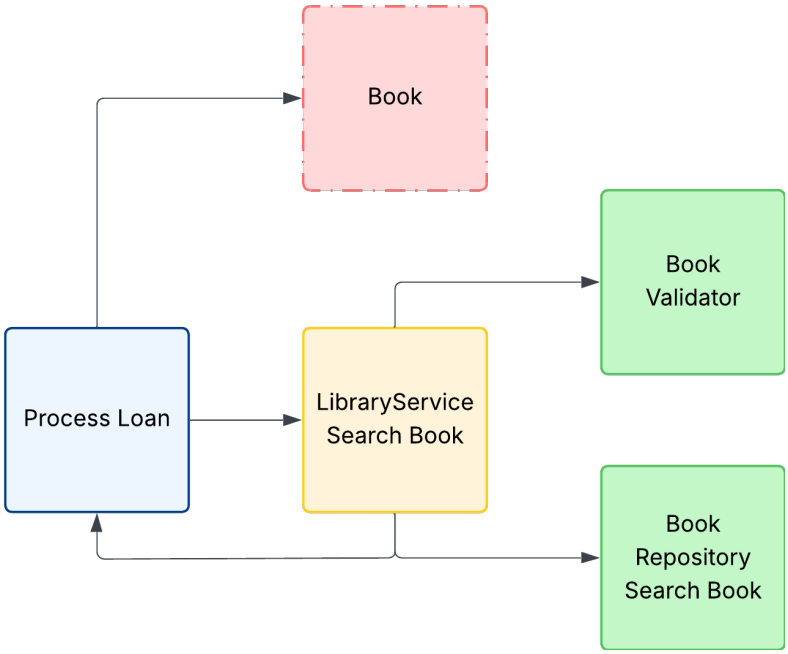
This is the core operational process of the system. The following describes the process loan workflow:

1. Navigate to the **Loans** menu item in the left navigation panel.
2. Click **Process Loan** which brings you to **processLoan.php**.
3. Enter the member ID in the search box and enter.
4. The system performs the following validations:
    a. Checks the member ID is valid.
    b. Checks the member is active.
    c. Checks the member has no overdue books.
    d. Checks the member has no outstanding fines.
5. If all checks pass, the member details are displayed on the UI.
6. Enter an ISBN in the book search box and enter.
7. The system performs the following validations:
    a. Checks the ISBN is valid.
    b. Checks the Book is not already in the cart.
    c. Checks the book is available.
    d. Checks the books in the cart do not exceed 5 books.
8. If all checks pass, the book details are displayed on the UI.
9. This process can be repeated for up to 5 books.
10. A loan summary is displayed showing the book and member details.
11. Click confirm checkout, which performs a final re-validation of the above and an additional check to ensure the member has not exceeded the maximum loan limit of 5 books.
12. If all checks pass:
    a. A loan is inserted into the Loans table.
    b. A loan item is inserted into the LoanItems table for each book.
    c. Each book is updated to show unavailable in the Books table.
13. The system redirects to **loanConfirmation.php** showing a loan confirmation and the loan details:
    a. Loan Date
    b. Due Date
    c. Member
    d. Books Loaned
14. From this screen, staff can click Print Receipt to generate a PDF receipt using the FPDF library and **receipt.php**.

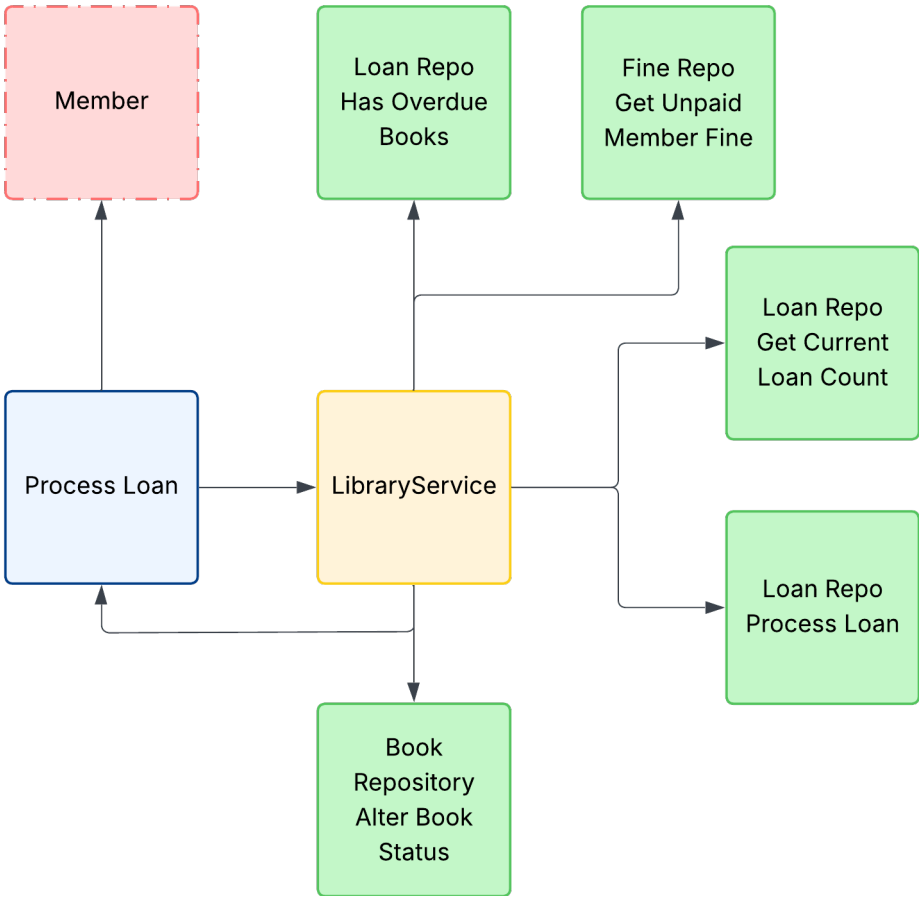
4.6.1.1. Process Loan – Search Member



4.6.1.2. Process Loan – Search Book



4.6.1.2. Process Loan – Transaction Process



5. System Testing

5.1 Add Book Testing

Animal Farm

George Orwell

Book about animals

978-0-141-03613-7

Fantasy

Secker and Warburg

17/08/1945

A

5.2. Add Member Testing

Andrew

Finnegan

31/04/1986

0872718493

Finnegandrew@gmail.com

1 Orchard Way

Fairview Heights

Killorglin

Kerry

V93H274

01/01/2020

A

END OF MANUAL

Library Management System

Server-Side Development

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INSTITUTION	Munster Technological University
YEAR	2026

SUBMITED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE SERVER-SIDE DEVELOPMENT FINAL PROJECT